

Design Your Own Digital Photo Frame

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This digital photo frame is designed using Arduino Uno. Smartphones with high-resolution cameras and enhanced features have motivated people to click photos of almost everything that they come across. It is not possible to frame all photos on a wall and, hence, the digital photo frame was invented in the 1990s, which would accept memory cards directly from digital cameras and display photos on the screen.

This proposed digital photo frame is quite inexpensive. You just need Arduino Uno, 8.89cm (3.5-inch) TFT colour LCD shield with 320×480 resolution and onboard micro-SD card slot. The main advantage of the shield is that it is compatible with Arduino Uno, Mega and other boards. The TFT LCD shield used in this project is shown in Fig. 1, and the connection diagram between Arduino Uno and the shield is given in Fig. 2.

Hardware library `MCUFRIEND_kbv` for the TFT LCD shield module can be downloaded from https://github.com/prenticedavid/MCUFRIEND_kbv

The library folder consists of several sample programs. Sample code `showBMP_kbv_Uno.ino` available in `MCUFRIEND_kv` library folder is a demo program to perform a slideshow of images on the screen, which is refined to `display.ino` for this project and made available along with this article.

You also need a graphics library, called Adafruit GFX library, for the proposed design. This can be downloaded from www.arduino-libraries.info/libraries/adafruit-gfx-library

Two limitations that were noticed with the demo code are:

1. Images with bitmap extensions (*.bmp) can only be displayed on the screen, whereas the proposed digital frame should cater to various other formats of images, such as *.jpg,

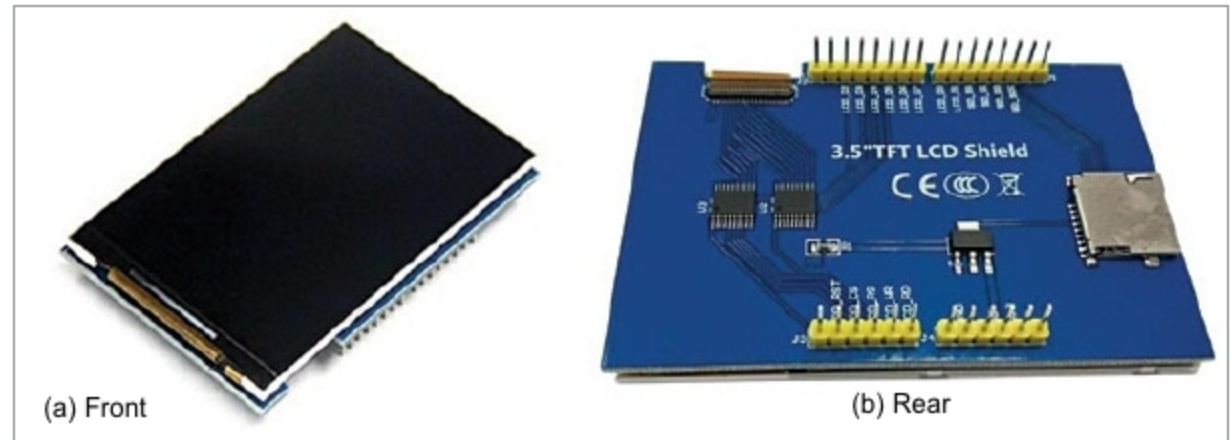


Fig. 1: TFT LCD shield used in the digital photo frame

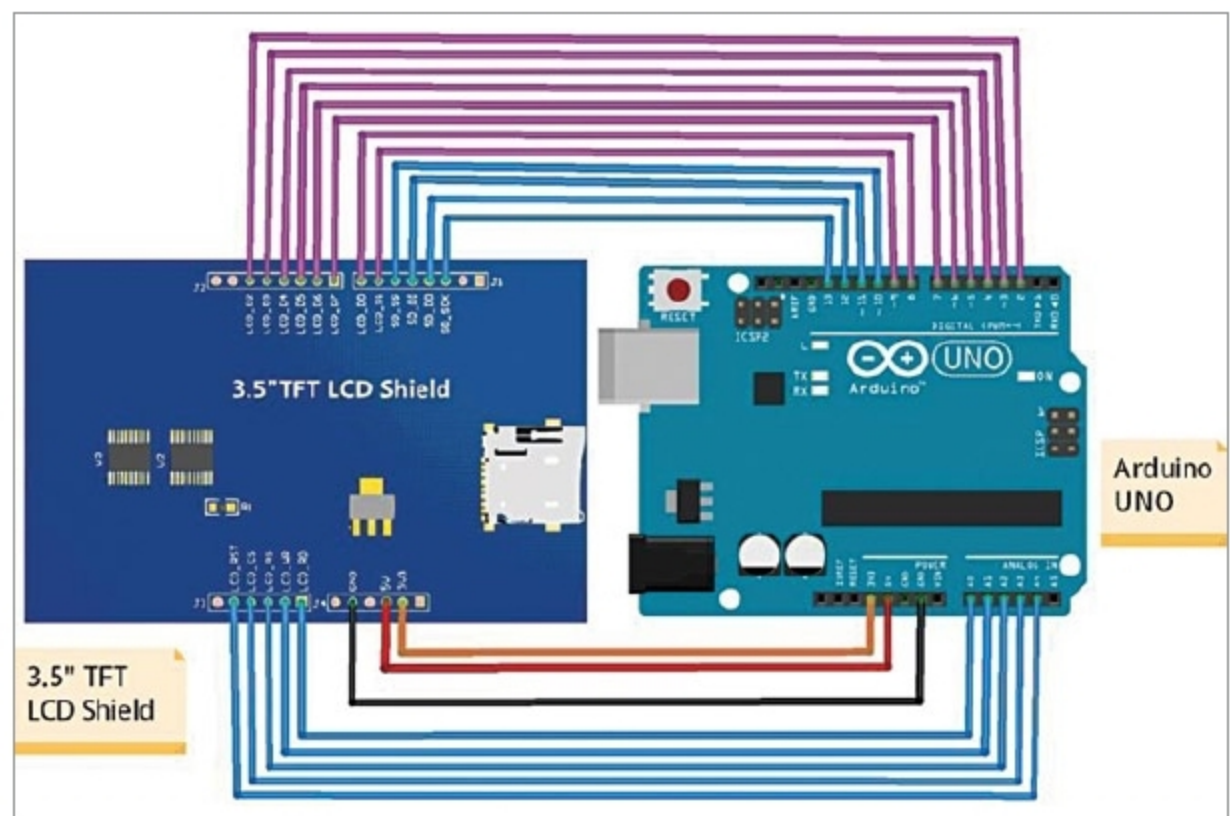


Fig. 2: Connection diagram between LCD shield and Arduino Uno

*.png, *.jpeg, *.tiff, etc.

2. Resolution of images that can be displayed is 320×480 pixels, whereas the proposed digital frame should cater to images of any resolution.

To take care of these two limitations, a dedicated software called Converter is designed using Python script (`converter.py`) to convert images of all other formats into *.bmp format with a resolution of 320×480 pixels for the proposed digital photo frame. You can convert format and size of each photo manually using Microsoft Paint, but this is a tedious job for multiple

photos. Converter software avoids the need of manual conversion.

To run Python script, you will need Pillow module, which comes inbuilt with some Python distributions. It can also be installed by running `pip install Pillow` in the terminal. Alternatively, you can directly run `converter.exe` file to start converting the files.

Snapshots of the sequence of software execution process are shown in Fig. 3. Once you select the source and destination folders, the software will identify the number of files and display the progress of the conversion